

RUSSIAN TORTOISE • TESTUDO HORSFIELDII

Keep your tortoise the right way for forty years.

A complete, fact-checked printable manual: enclosure size and why a tank is the wrong shape, UVB you can actually measure, a steppe grazer's real diet, the brumation decision, and the health pages for an animal that will outlast most of your furniture.

INSIDE

- ✓ Enclosure sizing, and why two credible sources give very different numbers
- ✓ A weed list, a grazing plan, and the never-feed list for an arid grazer
- ✓ The brumation decision, in full, with the protocol if you say yes
- ✓ Eight health pages, the legal position, and fillable owner and sitter sheets

37 pages

Intermediate

Print or tablet

Version 2.0 • Sep 2026



CONTENTS

What's inside

Every page is written to be used, not just read once.

GETTING STARTED

How to Use This Package	3
-------------------------	---

SECTION 01 · QUICK PROFILE

Quick Profile & Cost Overview	4
-------------------------------	---

SECTION 02 · FULL CARE GUIDE

Enclosure Size, Type & the Table Question	5
Temperature, Basking & Night Lows	6
UVB, Lighting & the Equipment That Controls It	7
Humidity, Substrate & the Moist Hide	8
Outdoor Housing & Escape-Proofing	9
Handling, Temperament & Why Males Live Alone	10
Diet: What a Steppe Grazer Actually Eats	11
Weeds, Grazing & Growing Your Own	12
Fruit, Protein, Supplements & the Never-Feed List	13
Common Mistakes & Enrichment	14
Sexing, Growth & Body Condition	15
Females, Eggs & Egg Binding	16

SECTION 03 · HEALTH & COMMON ISSUES

Health Red Flags & What to Tell the Vet	17
Metabolic Bone Disease & Pyramiding	18
Respiratory Infection & Herpesvirus	19
Shell Rot, Trauma & Abscesses	20
Parasites, the Fecal Test & Quarantine	21
Vitamin A Deficiency, Cloacoliths & Beak Overgrowth	22
Brumation: the Decision and the Protocol	23
Reading Droppings, Urates & Hydration	24

SECTION 04 · QUICK REFERENCE

The Law & the Four-Inch Rule	25
Setup Checklist & Targets	26
Emergency & Quick Targets Card	27

SECTION 05 · OWNER TOOLS

Budget & Shopping List	28
First 30 Days Checklist	29
Symptom Quick Reference	30
Daily, Weekly & Seasonal Routine	31
Power Outages, Travel & Transport	32
Pet-Sitter Sheet	33
Owner Log	34
Equipment, Supplement & Vet Log	35

REFERENCE

Glossary	36
Sources, Version History & About	37

START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

If you haven't bought a tortoise yet

Read pages 4 through 13 in order before buying anything. The enclosure on page 5 is the decision people get wrong most, because the thing sold as a tortoise tank is the wrong shape as well as the wrong size, and the UVB fixture on page 7 is the one you cannot substitute cheaply. Then use the Budget and Shopping List on page 28. Read page 25 too: the four-inch rule shapes what you can legally buy and from whom, and almost everyone repeats it wrongly.

If your tortoise is already set up

Audit against the Setup Checklist on page 26. Two things are worth checking today whatever else you do: the actual surface temperature of the basking spot with an infrared thermometer, and the age of the UVB bulb, because output falls away months before the light looks dim. Between them they cause most of what is on pages 18 and 19.

Print these first

- | | |
|--|---|
| ☐ Page 26: Setup Checklist & Targets | ☐ Page 33: Pet-Sitter Sheet |
| ☐ Page 27: Emergency & Quick Targets Card | ☐ Page 34: Owner Log |
| ☐ Page 30: Symptom Quick Reference | ☐ Page 35: Equipment, Supplement & Vet Log |
| ☐ Page 31: Daily, Weekly & Seasonal Routine | ☐ Page 13: the never-feed list, for the fridge |

WHY THIS PACKAGE EXISTS

Russian tortoise advice is unusually contradictory, because the animal is genuinely hardy and so a lot of substandard care produces a tortoise that survives for years while slowly deforming. The commonest failures, a glass tank, weak UVB, damp bedding, and too much food, all take years to show. Every figure here is cross-checked against veterinary and specialist tortoise sources, listed on page 37, and every temperature and dimension is given in both imperial and metric.

NOT A SUBSTITUTE FOR

A reptile veterinarian with tortoise experience. This is a husbandry reference for day-to-day care. Anything health-related is a professional call, not a page in this book. Find that vet before you need one, and write the number on page 27. A general small-animal clinic often cannot help with a tortoise, and this species lives long enough that you will need one eventually.

SECTION 01

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Testudo horsfieldii</i> . Also sold as the Horsfield's tortoise, steppe tortoise, or four-toed tortoise
Native range	Central Asia: Kazakhstan, Uzbekistan, Iran, Afghanistan, Pakistan and western China
Habitat	Semi-arid steppe, rocky desert and dry scrub. A burrowing animal, not a rainforest one
Adult size	5 to 10 in (13 to 25 cm) shell length; the largest verified specimens reach about 22 cm (8.7 in) straight carapace length. Females are usually larger than males
Captive lifespan	40 years or more, with individuals reaching 50 plus. Plan for succession of care, not just your own ownership
Diet type	Herbivore. High fiber, low protein, low sugar, high calcium. Page 11
Indoor enclosure	4×2 ft (120×60 cm) is the commonly quoted floor, 8×4 ft (240×120 cm) the modern species-reference target. Page 5 explains the gap
Enclosure type	Open-topped tortoise table, not a glass tank. This is a ventilation and behavior issue, not a preference
Basking surface	95 to 100°F (35 to 38°C), measured on the surface with an infrared gun
Cool end	70s°F (about 21 to 26°C). Tolerates night lows to 60°F (16°C) without supplemental heat
UVB	Linear T5 HO tube, target UVI 3.0 to 4.0 at the basking site. Page 7
Humidity	Low ambient, with a moist hide. Two credible sources disagree on the number. Page 8
Brumation	Physiologically programmed, 2 to 4 months, and optional indoors. Page 23
Social	House alone, or with care two females. Never two males. Page 10
Legal status	No state bans it; New Jersey requires a permit. Page 25. IUCN Vulnerable, CITES listed

Cost overview

Initial setup

\$400 to \$600 for a full indoor setup, of which \$125 to \$200 is the table alone. The tortoise itself is \$50 to \$400 depending on age and origin. Page 28.

Ongoing

\$500 to \$650 a year: food, substrate, UVB replacement and electricity, which rises in winter in a cold climate.

THE NUMBER THAT SHOULD DECIDE THIS

Forty years. A tortoise bought for a ten-year-old is likely to outlive that child's own children's interest in it. Rehoming a Russian tortoise is genuinely common and genuinely difficult, so the honest question before buying is not whether you can care for one now but who takes it in 2050. Write that person's name on page 27.

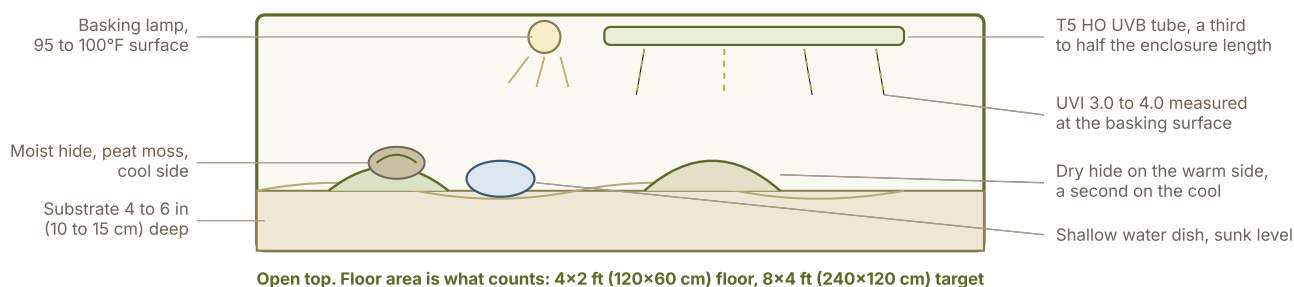
Enclosure size, type & the table question

Floor area, open top, and the reason a glass tank fails this species twice over.

Two credible sources give very different numbers and both are printed here rather than averaged. General care guidance, including our own, gives an indoor minimum of **4×2×2 ft (120×60×60 cm)** and describes a functional size closer to 7 ft by 3.5 ft. The ReptiFiles species care sheet gives **8×4 ft (240×120 cm), about 32 sq ft of floor**, with at least 10 sq ft (1 m²) more per additional tortoise. Treat 4×2 ft as the floor below which an adult is not adequately housed, and 8×4 ft as the target that a Russian tortoise will actually use, because this is one of the most active tortoises kept and it walks the whole of whatever it is given.

NOT A GLASS TANK

An open-topped tortoise table is strongly preferred, for two separate reasons. **Ventilation:** a closed glass vivarium traps humidity against an arid-adapted animal, which is a direct cause of respiratory infection (page 19). **Behavior:** a tortoise sees through glass but cannot reach what is beyond it, and the result is restless pacing along the wall for hours a day. Neither problem is fixed by buying a bigger tank.



Building or buying

A tortoise table is a shallow open box with solid sides, usually wood sealed against moisture, and it is easy to build and cheap to enlarge. Length matters more than width, and both matter more than height. Do not add a second level: this is a walking animal, not a climbing one, though it climbs better than it looks, which is the subject of page 9.

- **Walls 12 in (30 cm) or more**, solid rather than mesh, with nothing next to them a tortoise can use as a step. Corners are ladders; round them or block them.
- **A sealed, waterproof floor.** Damp substrate against bare wood rots it and grows mold. Pond liner or sealed plywood, with a lip so substrate cannot spill.
- **A genuinely cool end.** If the lamp warms the whole box, the box is too small, whatever the tape measure says.
- **Room for the lamps above it**, at the mounting distances on page 7, without a shelf or a lid in the light path.

HATCHLINGS DO NOT NEED A SMALLER ENCLOSURE

A common piece of advice is to house a young tortoise small so it can find its food and water. It is not necessary and it wastes money on a second enclosure. Give a hatchling the adult footprint with more cover: extra hides, more plants, and food and water in the same place every day. A young tortoise with places to hide uses a large space happily.

Temperature, basking & night lows

A genuine gradient, measured on the surface, in an unusually cold-hardy species.

A tortoise cannot make its own heat, so the enclosure has to offer a range and let the animal choose. What matters is a real gradient across the length of the table: a hot basking spot at one end and a genuinely cool end at the other. If the whole enclosure reads the same temperature, the animal has no way to regulate itself, and that is true whether the reading is too high or exactly right.

ZONE	TARGET
Basking surface	95 to 100°F (35 to 38°C), measured <i>on the surface</i> with an infrared temperature gun, not in the air
Warm end, ambient	Around 80 to 85°F (27 to 29°C)
Cool end	The 70s°F, roughly 21 to 26°C. This end has to actually be cool
Night	A drop to about 60°F (16°C) is fine and is normal for the species. No supplemental night heat is needed in most homes
Act below	Sustained below 60°F (16°C) outside an intentional brumation, which is where respiratory infection starts. Page 32
Brumation, if chosen	Around 40 to 55°F (4 to 13°C). Page 23

THIS IS ONE OF THE MOST COLD-TOLERANT TORTOISES KEPT

In the wild, Russian tortoises brumate in abandoned rodent burrows 20 to 50 cm (8 to 20 in) down, where the surrounding soil sits at 6 to 10°C (43 to 50°F) and the animal's own body temperature runs 5 to 11°C (41 to 52°F). Overnight temperatures in the +4 to +5°C range (39 to 41°F) are tolerated without harm. That hardiness is real, and it is exactly why owners get away with cold, damp setups for years before the animal shows what it has cost.

Getting the heat right

- **A halogen flood bulb is the right heat source** for a basking spot. It puts out the short-wavelength infrared that penetrates and warms an animal properly, the way sunlight does. A ceramic emitter warms air rather than the animal and is a night or ambient tool, not a basking one.
- **Raise or lower the lamp, do not chase wattage.** Distance is the adjustment. A dimming thermostat or a lamp dimmer on the basking bulb lets you hold a temperature through changing room conditions instead of guessing.
- **Never a heat mat under a tortoise.** A tortoise's plastron is not built to take belly heat, it does not register overheating from below, and burns are a real outcome.
- **Two thermometers, permanently.** A digital probe at the cool end and an infrared gun for the basking surface. Stick-on dial gauges are decorative.
- **The basking spot and the UVB tube go at the same end**, so the animal gets heat and UV together, which is how it works outdoors. Page 7.

HEAT IS THE EMERGENCY IN SUMMER, NOT WINTER

A cold-adapted steppe animal in a conservatory or a sunny room in July is in far more danger than the same animal in an unheated room in January. Outdoors, a pen with no permanent shade is a lethal enclosure. Indoors, check the enclosure's afternoon temperature rather than its morning one, at the hottest point of the year.

UVB, lighting & the equipment that controls it

The one piece of kit you cannot substitute, and the one that fails invisibly.

UVB drives vitamin D3 synthesis, which drives calcium absorption, which builds and maintains a shell. Without it a tortoise develops metabolic bone disease however much calcium is in the food (page 18). What makes it dangerous is that output falls away months before the bulb looks dim, so a fixture that is visibly working can be delivering nothing useful.

REQUIREMENT	DETAIL
Bulb type	A linear T5 HO tube in a reflective fixture. Arcadia 12% or Zoo Med ReptiSun 10.0 are the standard choices
Length	Roughly a third to half the enclosure length, mounted over the warm end alongside the basking lamp
Target	A UV index of 3.0 to 4.0 at the basking area, measured with a UV index meter
Mounting distance	With mesh in the path, the tortoise's back sits 14 to 16 in (35 to 40 cm) below the tube. With no mesh, 17 to 20 in (43 to 50 cm)
Replacement	Every 12 months, on the calendar, not when it stops lighting. Write the date on the bulb and on page 35
Visible light	A bright 6500K daylight lamp alongside. Tortoises are diurnal and a dim enclosure suppresses activity and appetite
Photoperiod	12 hours on, 12 off, on a timer. Lengthen toward 14 in summer and shorten toward 10 in winter to track the seasons

THREE WAYS TO DELIVER NO UVB WHILE BELIEVING YOU ARE

A coil or compact bulb. They concentrate output into a narrow cone, have been associated with eye and skin irritation, and are not adequate for a species that needs a UVI of 3 to 4 across a basking area.

Glass or plastic in the path. Glass removes essentially all of it, which is also why a sunny window is not sunlight to a tortoise. Mesh costs a substantial fraction, which is what the two mounting distances above are for.

A bulb over a year old. Output decays continuously from the day it is switched on.

The equipment that controls the rest

- **A timer** for the lights, so the photoperiod holds whether or not you are home. The cheapest item on the list.
- **A dimming thermostat or lamp dimmer** on the basking bulb, holding the surface temperature through changing room conditions. An on/off thermostat is the wrong controller for a basking lamp, because it flickers the light.
- **An infrared temperature gun** for the basking surface. Under \$25, and it settles what the other instruments cannot.
- **A digital probe thermometer** at the cool end, plus a hygrometer. Page 8.
- **A UV index meter**, optional and expensive, and the only way to know rather than assume. Borrow one if a local keeper has it.

SUNLIGHT BEATS EVERY BULB

Unfiltered natural sunlight delivers UVB no lamp matches, so a few supervised hours outdoors in warm weather is worth more than any equipment upgrade. Page 9 covers doing it safely. It does not replace an indoor fixture: the weather does not cooperate for most of the year.

Humidity, substrate & the moist hide

An arid species that still needs somewhere damp, and an open disagreement about the number.

TWO CREDIBLE SOURCES, TWO DIFFERENT FIGURES

General care guidance for this species, ours included, gives ambient humidity of **30 to 50%** and warns that high ambient humidity across the whole enclosure is a real cause of respiratory infection. The ReptiFiles species care sheet gives **40 to 75%, lower during the day and higher at night**, following what the animal experiences in a burrow. Both agree on the shape of the answer even where the numbers differ, and the shape is what matters: **a dry, well-ventilated open surface with a genuinely humid microclimate available underneath and inside a hide**. A tortoise in the wild spends most of its life in a burrow that is far damper than the steppe above it. Build both, and let the animal choose.

In practice: keep the open surface dry and the air moving, keep the deeper substrate slightly damp rather than bone dry, and provide at least one hide packed with moist sphagnum or peat moss on the cool side, remoistened every few days. That combination gives hydration and clean shedding without a permanently damp enclosure, which is what actually causes the respiratory and shell problems on pages 19 and 20.

Substrate

USE	NOTES
50/50 topsoil and coconut coir	The standard mix. Holds a burrow shape, holds a little moisture at depth, and is cheap in bulk. Buy topsoil with no added fertilizer, pesticide or wetting agent
Sandy loam, or a topsoil and play sand blend	Closer to the native substrate. A minority of sand in a soil mix is fine and useful
Depth	At least 4 in (10 cm), which is the minimum for the animal to dig a pallet, and 6 in (15 cm) or more in at least part of the enclosure is better
Avoid: pure sand	Linked to mobility problems and impaction, and it does not hold a burrow
Avoid: heavy wet clay soils	Wrong for a desert species and slow to dry
Avoid: newspaper or rabbit pellets alone	No digging, no thermal buffering, and nothing the animal can do with them
Avoid: cedar and pine shavings, walnut shell, calcium sand	Respiratory irritation, impaction risk, and none of them behave like soil

DIGGING IS NOT DECORATION

Burrowing is how this species buffers temperature and humidity, and a substrate too shallow to dig into removes a behavior the animal actively needs rather than merely enjoys. If the main substrate cannot go deep enough, build a dedicated dig box: a plastic tub of damp soil mix sunk into the enclosure works, and it will be used constantly.

Soaking and water

Keep a shallow water dish available at all times, sunk level with the substrate so the animal can walk in and out and cannot tip. Change it daily; tortoises defecate in water routinely. Beyond that, **soak the tortoise in shallow warm water for 20 to 30 minutes weekly**, deeper for a young or newly acquired animal. The water should reach the bottom of the shell and no higher. Most tortoises drink and pass urates during a soak, which is also when you get the best look at both. Page 24.

Outdoor housing & escape-proofing

The best enclosure this species can have, and the one it will try hardest to leave.

Whenever climate and security allow, outdoor housing is the richest environment available to this species: unfiltered sunlight, real soil to dig, variable terrain, weather, and edible plants growing where the animal walks. Many long-term keepers treat an outdoor pen as the standard and the indoor table as the winter arrangement. A supervised hour on unsprayed grass beats any equipment purchase.

Building a pen

ELEMENT	WHAT IT NEEDS
Walls	Solid rather than mesh, so the tortoise cannot see through and pace, and at least 12 in (30 cm) above ground with nothing climbable against them
Buried barrier	Walls dug in at least 12 in (30 cm), and ideally 30 cm of concrete, paving or wire mesh set into the ground. This species digs persistently and will tunnel out of a pen that looks secure
Shade	Permanent, in the hottest part of the day, from a structure rather than from a plant that may be cut back. Heat is the real outdoor danger
Shelter	A dry insulated house the animal can retreat into for cold or wet weather, with a low entrance and bedding of hay or dry leaves
Substrate	Diggable soil, not lawn over hardpan, with a dig area at least 6 in (15 cm) deep
Planting	Edible weeds sown across the pen so the animal grazes as it walks. Page 12
Cover overhead	Wire or netting where birds, cats, dogs, rats or foxes are a possibility. A small tortoise is prey to more animals than owners expect

FIVE WAYS AN OUTDOOR TORTOISE DIES

Heat. A pen with no permanent shade at 2 pm in July.

A dog. Shell trauma from a household dog is one of the commonest serious injuries in pet tortoises and it is very often the family's own animal.

Escape. Digging under, or climbing a corner. Corners are ladders, so round them off or block them.

Drowning. A pond, a pool, a deep water bowl, or an upended tortoise in a shallow dish it cannot right itself out of.

Pesticide. Sprayed lawn, treated soil, slug pellets, or a neighbor's overspray. Nothing goes on a pen and nothing grazed came off treated ground.

When outdoors is not an option

Indoor keeping is genuinely viable for this species for its entire life, which is the main advantage it has over the larger tortoises. It requires the UVB fixture on page 7 to be right rather than approximately right, and it makes the enclosure size on page 5 matter more, because there is no garden making up the difference. A weekly supervised session outside in warm weather bridges the two arrangements and is worth arranging even for a committed indoor keeper.

TAKE THE TEMPERATURE OUT THERE TOO

Pen surfaces reach temperatures a lawn does not. Check a paving slab, a stone and the shaded soil with the infrared gun on the hottest afternoon of the year before you trust the pen. If a surface is too hot for your hand, it is too hot for a plastron.

Handling, temperament & why males live alone

An observation animal with a personality, not a lap animal.

Russian tortoises are active, curious and genuinely responsive to their keepers: many approach confidently, recognize the person who brings food, and rearrange their enclosure overnight. None of that makes them a handling pet. Being picked up is a predator experience for an animal whose entire defense is staying on the ground inside its shell, and over-handling causes real, measurable stress. Keep physical contact purposeful, brief and infrequent, and get the relationship from watching and from feeding instead.

Picking one up, when you have to

- Approach from the front or the side where it can see you, and pause. A hand descending from above is a bird.
- Support the whole shell from underneath with both hands, keeping the animal level and low to the ground.
- Never tip it vertically or onto its back. That compresses the lungs against the organs and is genuinely dangerous, not just unpleasant.
- Never lift by a leg or the front of the shell, and never pull a retracted head or limb.
- Sit on the floor. A dropped tortoise cracks, and a cracked shell is a page 20 problem.
- Wash your hands before and after. Reptiles carry *Salmonella* as normal gut flora, and it is the reason a tortoise is a poor pet for a household with children under five or anyone immunocompromised.

SALMONELLA, PLAINLY

Healthy tortoises shed *Salmonella* without being ill themselves. The route to people is hands, then face or food. Wash hands after every contact with the animal, its enclosure, its water dish or its substrate. Do not clean tortoise equipment in a kitchen sink or a bath used by people. Do not kiss the tortoise. This is the reason behind the federal four-inch rule on page 25, and it is a real risk rather than a regulatory relic.

Housing more than one

House alone by default. This is a solitary species that does not benefit from company the way a social mammal does, and an owner adding a second tortoise for the first one's sake is usually creating a problem. If you do keep more than one:

- **Never two males.** Males are genuinely territorial and will ram, bite legs and flip one another, sometimes fatally. This is not a personality question and it does not resolve with more space.
- **Two females can work**, with caution, in a large enclosure with several hides, several feeding stations and sight breaks between them. Watch for one animal monopolizing the basking spot.
- **A male with a single female is a problem for the female**, who gets pursued and mounted relentlessly. A group of several females to one male spreads that, which is a breeding arrangement rather than a pet one.
- **Never mix species.** Different husbandry needs, and pathogens that one species carries harmlessly can be fatal to another. Page 21.

BETTER THAN THEY LOOK, AT ESCAPING

Despite the reputation, Russian tortoises climb and dig well and are persistent about both. They will scale a corner, push under a barrier, or work at the same weak point for weeks. Assume any enclosure is escapable until it has held for a season, and check the perimeter as a routine job rather than after the first escape.

Diet: what a steppe grazer actually eats

High fiber, low protein, low sugar, high calcium. And less of it than you think.

The dietary rule for *Testudo* species is short: **low protein, low carbohydrate, low sugar, high calcium, high fiber**, and that combination is the primary key to deformity-free growth. In the wild a Russian tortoise eats from a range of over 80 plant species across 27 families, almost entirely weeds, leaves and flowers. It is a grazer of tough, fibrous, low-nutrient plants, and every problem on page 18 comes from feeding it something richer.

THE QUANTITY NOBODY MENTIONS

In nature this species typically feeds on no more than about **ninety days out of three hundred and sixty five**, and even then spends roughly fifteen minutes grazing a day. Captive tortoises are fed every day, generously, on food far richer than steppe weeds, which is why overfeeding is the single most common husbandry error in this species and why pyramiding is so widespread. Feeding less, and feeding coarser, is closer to correct than most keepers are comfortable with.

What the diet is made of

COMPONENT	SHARE	NOTES
Edible weeds and wild greens	THE CORE	Dandelion leaves and flowers, plantain weed, clover, sow thistle, hawkbit, mallow, chickweed, vetch. Page 12
Grass hay	ALWAYS AVAILABLE	Timothy, orchard or meadow hay, offered loose. Fiber and a fallback, and a tortoise will graze it steadily
Fibrous leafy greens	SUPPLEMENT	Endive, escarole, radicchio, mustard greens, collard greens, romaine and other dark leaf lettuces, turnip greens, watercress, arugula
Flowers	REGULARLY	Hibiscus, nasturtium, rose petals, pansy, dandelion. Unsprayed
Spinach, beet greens, chard, parsley, kale	ROTATE, SMALL	High in oxalate or calcium and fine occasionally, poor as staples because oxalate binds calcium
Commercial tortoise pellets	NOT PRIMARY	Grass-based rehydrated pellets can supplement a winter diet. They should never be the base, and mixed pellets with fruit or grain in them are worse than nothing
Fruit, protein, anything animal	NEVER	Page 13

How to feed

- **Scatter feed rather than using a bowl.** Put greens in several places so the animal has to walk and search, which is closer to grazing and doubles as enrichment. Page 14.
- **Feed on a flat stone or a tile** where you do scatter feed, so soil and substrate are not swallowed with every mouthful.
- **Adults: feed daily, in a portion roughly the size of the tortoise's shell**, and expect it to be eaten within a few hours rather than picked at all day. A fasting day once a week does an adult no harm at all and reflects how the animal eats in the wild.
- **Hatchlings and juveniles eat daily and more**, with no fasting days, because they are building shell.
- **Remove uneaten fresh food** the same day. Hay can stay.

Weeds, grazing & growing your own

The cheapest and best food for this species grows in the cracks of a pavement.

Supermarket greens are bred to be tender, watery and mild, close to the opposite of what a steppe grazer evolved to eat. Weeds are tougher, higher in fiber and minerals, more varied, and free. A keeper who learns six weeds by sight has better food than one spending \$200 a year on salad, and the variety matters: the wild diet runs to over 80 plant species.

WEED	HOW TO RECOGNIZE IT, AND NOTES
Dandelion	The staple. Deeply toothed leaves in a flat rosette, hollow stem with milky sap, single yellow flower. Leaves and flowers both, in quantity. High in calcium
Plantain (ribwort and broadleaf)	Ribbed parallel-veined leaves in a rosette, with a bare flower spike. Grows in every lawn and path edge. Excellent, and tortoises usually take to it immediately
White and red clover	Three leaflets with a pale crescent, ball-shaped flower. Good, but higher in protein than most weeds, so a component rather than a base
Sow thistle	Tall, hollow stem with milky sap and clasping leaves, small dandelion-like flowers. A favorite. Do not confuse with prickly lettuce, which is also fine
Hawkbit, cat's ear, chickweed	Hawkbit and cat's ear are dandelion lookalikes with hairy leaves and branched stems, and all are fine to feed, which is why the lookalike problem hardly matters. Chickweed is a low sprawling mat with white star flowers: fine, but watery
Mallow	Rounded lobed leaves, pink-purple flowers. Leaves and flowers both, and it is a good hot-weather grower
Vetch, hawksbeard, bramble leaf, hibiscus leaf	All safe and all worth adding for variety

WHERE YOU MUST NOT PICK

Anywhere sprayed with herbicide, pesticide or fertilizer, including a neighbor's overspray. Within a few meters of a road, where exhaust residue and salt accumulate. Anywhere dogs are walked. Any verge a council or contractor maintains. Farmland margins. And never pick something you cannot name: buttercup, ragwort, foxglove, hemlock, ivy, deadly nightshade and daffodil all grow alongside the good weeds and several of them are lethal. If you are not certain, leave it.

Growing your own

A seed tray or two of a tortoise forage mix, sown in rotation, turns cut greens into living plant the animal crops itself, which is as close as a captive setup gets to grazing. Three trays on a windowsill on a two-week rotation keeps a single tortoise supplied through most of the year. Outdoors, sow forage mix straight into the pen and let the animal graze as it walks; that is the single biggest quality-of-life change available for this species. A pot of dandelion and plantain in a sunny corner covers the winter.

DRYING WEEDS FOR WINTER

Dandelion, plantain and clover all dry well. Cut on a dry day, spread in a single layer somewhere warm and airy out of direct sun until crisp, then store in a paper bag rather than plastic. Crumbled into hay it turns a dull winter diet into a varied one, and it costs an afternoon in June.

Fruit, protein, supplements & the never-feed list

The short page that prevents most of the health section.

NEVER FEED A RUSSIAN TORTOISE

Any animal protein. Dog or cat food, boiled egg, cheese, meat, insects, mealworms. Excess protein drives pyramiding and burdens the kidneys, and none of it belongs in a steppe herbivore.

Fruit. Fruits are inappropriate for arid-habitat species: the sugar disrupts gut flora, causes loose droppings and encourages the wrong fermentation. If you feed any at all it belongs under 5% of the diet, and zero is a defensible answer for this species.

Bread, pasta, grain, cereal, cat treats, "tortoise" biscuits.

Iceberg lettuce. Almost no nutrition, and linked to vitamin A deficiency in tortoises fed on it. Page 22.

Toxic plants: avocado, rhubarb leaves, buttercup, ragwort, foxglove, daffodil and other bulbs, ivy, hemlock, nightshade, azalea, oleander, yew.

Anything from treated ground. Page 12.

Why protein and fruit matter so much here

A tortoise's shell grows continuously, in layers, at whatever rate the diet allows. Fed richly, it grows fast and unevenly, and the scutes lift into the raised cone shapes called pyramiding, which is permanent (page 18). Fed on tough, fibrous, low-nutrient plants, it grows slowly and smoothly, which is what it does in the wild. The commonest owner instinct, that a growing animal should be fed generously and well, is the exact instinct that deforms this one.

Supplements

SUPPLEMENT	SCHEDULE
Calcium without D3	A light dusting on food at most feedings. This is the daily one
Calcium with D3	Twice a week at most, and less or none for a tortoise with strong UVB and outdoor time, since D3 is what the UVB is for. Never at every feeding
Reptile multivitamin	Once a week or so, herbivore-appropriate, with no added phosphorus
Cuttlebone	Left in the enclosure permanently, for the tortoise to gnaw at will. Also helps keep the beak in shape. Page 22

Supplement schedules vary by keeper and by how much natural sun the animal gets, so treat this as the starting point and ask your reptile vet to review it at the annual exam. A tortoise living outdoors in summer needs less added D3 than one under a bulb in January.

DO NOT DUST WITH D3 AND THEN ADD A D3 MULTIVITAMIN

Vitamin D3 is fat-soluble and accumulates, and oversupplementation causes its own problems. Read both labels, count the D3 sources you are actually giving, and remember that a strong UVB fixture is one of them.

WATER IS A SUPPLEMENT TOO

A shallow dish always available, changed daily, plus a weekly 20 to 30 minute soak in shallow warm water. Hydration does as much for smooth shell growth as any powder, and a dehydrated tortoise is the one that pyramids and the one that forms the urate blockages on page 22.

Common mistakes & enrichment

Ten things that go wrong most, and what an interesting enclosure looks like.

THE MISTAKE	WHAT TO DO INSTEAD
A glass tank instead of an open table	Ventilation and pacing, both. Page 5
Buying for the size it is, not the size it becomes	Buy the adult enclosure now. Page 5
A coil UVB bulb, or one over a year old	Linear T5 HO, replaced on the calendar. Page 7
Measuring air temperature instead of the basking surface	Infrared gun, on the surface. Page 6
Misting the whole enclosure to raise humidity	Dry surface, moist hide. Page 8
Substrate too shallow to dig into	4 to 6 in (10 to 15 cm), or a dig box. Page 8
Feeding supermarket salad and fruit daily	Weeds, hay and fibrous greens. Pages 11 and 12
Overfeeding a growing tortoise "to help it grow"	That is what causes pyramiding. Page 13
Two males in one enclosure	Never. Page 10
No fecal test on a newly acquired animal	Do it in the first month, symptoms or not. Page 21

What enrichment means for a tortoise

The behaviors this species is built for are walking, grazing, digging and thermoregulating, so enrichment is mostly a matter of building an enclosure those can happen in rather than adding objects to it. A Russian tortoise explores its whole territory every day, pushes things over, and rearranges what it can move. An enclosure that looks tidy the next morning is usually one that is too small or too empty.

Terrain and structure

- Flat stones, cork flats, gentle slopes and low platforms, which do get used rather than walked past.
- Large hollow logs and visual breaks, so the space is not one flat open plain.
- Multiple hides facing different directions, one warm, one cool and moist.
- A deep dig area, or a dedicated dig box.

Routine and change

- Scatter feeding in new places daily or every other day.
- Living plants to crop rather than cut greens in a dish. Page 12.
- Rearrange non-critical decor every one to four weeks, without moving the hides themselves.
- Supervised outdoor time whenever the weather allows. Page 9.

LEAVE THE HIDES WHERE THEY ARE

Rearranging the scenery is enrichment; moving the shelter is stress. A tortoise builds a mental map of where safety is, and shifting the hide means it has to find safety again from scratch. Change the route, not the destination.

TWO THINGS THAT LOOK LIKE ENRICHMENT AND ARE NOT

A mirror or a reflective surface. A male reads a reflection as a rival and will ram it. **A companion added for company.** This is a solitary species, and page 10 is the reason.

Sexing, growth & body condition

Straightforward to sex once mature, and worth weighing every month for forty years.

Sexing

Russian tortoises are not reliably sexable as hatchlings or small juveniles; the differences appear as the animal matures, generally somewhere past about 4 in (10 cm) of shell. Once they do appear they are clear.

FEATURE	MALE	FEMALE
Tail	Much longer and thicker, often carried tucked to one side. This is the reliable one	Short and blunt
Vent position	Further out along the tail, past the shell margin	Close to the base of the tail
Plastron	Barely convex, sometimes very slightly dished	Flat
Carapace shape	Generally flatter	Generally higher domed
Size	Usually smaller	Usually larger
Behavior	Ramming, biting, head bobbing, chasing. Page 10	Digging test nests in season, whether or not a male exists. Page 16

Growth, and why slow is right

There is no growth chart for this species that is worth printing, because a healthy Russian tortoise grows at whatever rate its diet and its season allow, and the wild pattern is slow and interrupted. Fast growth is a warning rather than an achievement: a tortoise that gains rapidly on a rich diet grows the raised, cone-shaped scutes of pyramiding, which never flatten out again. What you are watching for is **smooth** growth, with new keratin appearing as flat pale rings around the edge of each scute and the surface of the shell staying even.

THE THREE THINGS TO CHECK MONTHLY

Weight, on a digital kitchen scale in grams. Write it down. Trends over years are what this record is for, and the reason a tortoise's owner log runs for decades.

Shell surface. Run a finger across the scutes. Smooth is right; a growing ridge at the scute margins is the beginning of pyramiding and it is a signal to reduce food and increase hydration and exercise now, while it still matters.

Shell rigidity. Press very gently at the rear edge of the carapace and along the plastron. A healthy adult shell does not flex. Any give at all in an animal past hatchling size is metabolic bone disease until a vet says otherwise. Page 18.

Body condition

A tortoise in good condition feels heavy for its size when lifted, the skin around the limbs looks full rather than loose, and the animal can withdraw its head and legs fully without the limbs looking like they barely fit. Signs of poor condition are a light, hollow feel, loose wrinkled skin at the limb sockets, sunken eyes, and lethargy. Because so much of the animal is shell, weight is a better measure than appearance, and the jump-to-conclusion figure worth knowing is that **any sudden weight loss in a tortoise is significant**, since these animals lose weight slowly by nature.

Females, eggs & egg binding

A female with no male in the house can still produce eggs, and she still has to pass them.

A mature female develops and lays eggs on her own hormonal schedule, whether or not she has ever met a male. Those eggs are infertile, but the physical process is identical and so is the risk if she cannot complete it. Every owner of a female should know what a gravid female looks like and what to do when the laying does not happen.

Signs a female is gravid

- **Restlessness and pacing**, often out of character, and walking the perimeter repeatedly.
- **Digging test nests**. Repeated shallow excavations, often abandoned, usually in a warm spot.
- **Reduced appetite** as the eggs take up room.
- **A firm, rounded feel** if a vet palpates the soft area in front of the hind legs. Do not attempt this yourself; pressing on eggs can break them.

PROVIDE A NEST SITE BEFORE SHE NEEDS ONE

A female with nowhere acceptable to dig will hold eggs, and holding eggs is how a straightforward event becomes an emergency. Give her a deep area of slightly damp substrate, at least as deep as the shell is long and ideally 8 to 12 in (20 to 30 cm), in a warm and undisturbed corner. Outdoors, a mound of soil in a sunny spot works. She may reject several sites before choosing one, and being watched puts many females off entirely.

EGG BINDING, OR DYSTOCIA

A female straining, restless for days, digging obsessively and then stopping, going off food, becoming lethargic, or producing fluid or blood from the vent is a same-day veterinary case. So is a female you know is gravid who has been trying to nest for more than a couple of days without laying. Left alone this ends in a ruptured oviduct, peritonitis or death. Treatment ranges from correcting the nest site and giving fluids and calcium, through hormonal induction, to surgical removal of the eggs. It is not something to wait out.

What to do with the eggs

With no male, they are infertile and can be discarded once she has finished and covered the nest. Let her finish: a female interrupted mid-clutch may retain the remainder. If a male has had access they may be fertile, and incubating them is a decision to produce more forty-year animals into a market where rehoming is already hard.

CALCIUM IS WHAT THIS COSTS HER

Producing a clutch draws heavily on a female's calcium reserves, which is the practical link between this page and page 18. A laying female needs her calcium supplementation kept up, a cuttlebone permanently available, and her UVB fixture in date. A female laying repeatedly on a marginal setup is the classic route to metabolic bone disease in an adult that was fine for years.

Health red flags & what to tell the vet

A hardy animal that hides decline for years, then declines quickly.

Almost every health problem here traces back to a handful of husbandry basics, and the pattern is worth learning as a pattern: weak UVB and low calcium cause metabolic bone disease; cold, damp or drafty conditions cause respiratory infection; overfeeding and poor hydration cause pyramiding; an unscreened new animal brings in parasites and viruses.

SAME DAY, NO WAITING

Open-mouth breathing, gasping, or extending the neck to breathe.

Bubbles or mucus at the nose or mouth, or discharge from the eyes.

Wheezing, clicking or any audible breathing.

Straining while vocalizing, which in a tortoise means a blockage. Page 22.

A female straining or unable to lay. Page 16.

Any crack, puncture or bleeding in the shell, particularly after a dog. Page 20.

Prolapse of anything from the vent.

A tortoise that will not right itself when flipped, or cannot support its own weight.

Sudden collapse or unresponsiveness after a hot day.

Within a few days

- Not eating for more than a few days outside brumation
- Any weight loss, which in this species is always significant
- Soft or flexible shell, or a rear shell margin that gives
- Swelling at the eyelids or beside the head, behind the eyes
- Soft, discolored or foul-smelling patches on the shell
- Diarrhea, or droppings that have changed persistently
- Lethargy, or a tortoise that has stopped walking its territory
- Overgrown beak, or difficulty biting through food

Targets that fix most of it

SETTING	TARGET
Basking surface	95 to 100°F (35 to 38°C)
Cool end	The 70s°F (21 to 26°C); night to 60°F (16°C) is fine
UVB	T5 HO tube, UVI 3.0 to 4.0 at the basking site, replaced every 12 months
Humidity	Dry ventilated surface, moist hide available. Page 8
Diet	Weeds and hay, low protein, no fruit. Pages 11 and 13
Soak	20 to 30 minutes in shallow warm water, weekly

WHAT TO HAVE READY WHEN YOU CALL

Current weight and the weight from a month ago. Basking surface and cool-end readings, actual rather than what the equipment is set to. The age of the UVB bulb. What it eats, in proportions. When it last ate, drank and passed droppings and urates. Whether it has been outdoors, and whether another reptile has been in the house. Photograph the enclosure, the shell top and bottom, and the last droppings.

Metabolic bone disease & pyramiding

The two conditions that are written permanently into the shell.

Metabolic bone disease

MBD is among the most common conditions in pet tortoises, and it comes from an imbalance of calcium and phosphorus caused by an improper diet, a lack of adequate ultraviolet light, or inadequate protein intake. Without usable D3 the animal cannot absorb calcium regardless of how much is in the food, so it draws calcium out of its own skeleton and shell to keep blood levels workable.

SIGNS

Missshapen shell. Deformed leg bones. Slow growth. A shell that stays flimsy beyond about six months of age. Weak limbs, an inability to lift the body clear of the ground when walking, or a wobbling gait. In severe cases, fractures that occur with no real trauma. Advanced cases can leave permanently misshapen bones or a permanent shell deformity even after treatment.

Prevention is the whole story: a T5 HO UVB tube at UVI 3.0 to 4.0 replaced yearly, calcium without D3 on most meals, a cuttlebone always available, and a high-calcium low-phosphorus plant diet. Treatment is veterinary, involving injectable calcium and often vitamin D alongside correcting the husbandry. Do not treat a suspected case with supplements alone: calcium given without addressing the D3 pathway is not absorbed.

Pyramiding

Pyramiding is abnormal raised growth of the scutes, so the carapace carries individual sections that look like small pyramids instead of lying flat. It is associated with suboptimal temperature and humidity and with imbalances of calcium, phosphorus and protein, and in practice the drivers are excess dietary protein, overfeeding, and insufficient hydration or humidity. It is entirely preventable and **completely irreversible**: once the scutes are raised they stay raised for the animal's life. A tortoise with mild pyramiding that is otherwise well cared for can live a normal lifespan, so it is a permanent record of past husbandry rather than an ongoing illness. Severe pyramiding does restrict movement and lung capacity.

DRIVER	FIX
Too much food, too often	Adult portion the size of the shell, daily, with a fasting day weekly. Page 11
Too much protein: pellets, legumes, any animal food	Weeds and hay. Page 13
Chronic mild dehydration	Weekly soaks, water always available, a moist hide. Page 8
An enclosure that is dry everywhere, including the burrow	Dry surface, damp depth. Page 8
No exercise, in an enclosure too small to walk	Page 5 and page 9

READING THE SHELL AS A HISTORY

A shell records the conditions it grew in. Smooth centers with raised outer rings means the animal grew well early and badly later; the reverse means a bad start that has been corrected. Either way what you can change is the next ring, so treat existing pyramiding as information rather than as something to fix, and check the growing margin monthly as described on page 15.

Respiratory infection & herpesvirus

The urgent one on this list, and the one that can take a whole collection.

Respiratory tract infection

Caused by *Mycoplasma*, other bacteria, or viruses, and often secondary to vitamin A deficiency. The husbandry triggers are chronically low temperature, excess humidity across the whole enclosure, and drafts, which is why the ventilation argument on page 5 and the humidity argument on page 8 are health arguments rather than comfort ones.

SIGNS, AND THIS DOES NOT RESOLVE ON ITS OWN

Excess mucus in the mouth, seen as bubbles at the mouth, nose or eyes. Nasal discharge. Lethargy. Loss of appetite. Wheezing. Extending the neck to breathe. Open-mouth breathing or gasping. Any of these is a veterinary appointment now, not a wait-and-see, and treatment is antibiotics given orally, by injection or as nose drops. Severe cases need hospitalization with fluid therapy and assisted feeding.

While you arrange care, raise the warm end toward the top of the normal range, make sure the animal is hydrated, and remove any source of damp or draft. Do not raise the humidity, which is a common and counterproductive instinct.

Herpesvirus, and why quarantine is not optional

THE REASON A NEW TORTOISE NEVER JOINS AN EXISTING ONE

Testudo tortoises, this species included, are susceptible to epidemics of viral disease and should be treated as potential high-risk carriers of herpes-type organisms. An infected animal is highly contagious and can appear healthy for long periods; in a group with no quarantine, mortality can approach 100%. Signs when they appear include a runny nose, swelling, a white or yellow plaque in the mouth and on the tongue, difficulty swallowing, lethargy and neurological signs. There is no cure. Treatment is supportive and a survivor remains a carrier for life.

What follows from that is simple and absolute. A new tortoise is quarantined in a separate room, on separate equipment, for months rather than weeks, and is screened by a vet before it goes anywhere near another tortoise. Page 21 sets out the full protocol. Never buy a tortoise on impulse from a table at an expo where animals from many sources have been sitting together, and never accept a tortoise from a household whose other animals you have not asked about.

Ranavirus, and mycoplasma

Ranavirus infections also occur in chelonians and, like herpesvirus, often prove fatal. *Mycoplasma* causes the chronic upper respiratory tract disease familiar to tortoise keepers: a persistent runny nose that comes and goes for years, worsens under stress, and is very difficult to clear. All three are arguments for the same two habits: quarantine properly, and never mix species or sources.

A RUNNY NOSE IS NEVER JUST A RUNNY NOSE IN A TORTOISE

Owners describe it as a cold and wait. There is no tortoise equivalent of a cold that resolves by itself. Persistent nasal discharge is either an infection, a chronic mycoplasmal disease, or vitamin A deficiency, and all three need a vet and none improve with time.

Shell rot, trauma & abscesses

The shell is living bone with a skin over it, and it gets infected like any other tissue.

Shell infection, or shell rot

Shell infections, also called shell rot and historically SCUD, or septicemic cutaneous ulcerative disease, are caused by bacteria, fungi or parasites entering through trauma: a scrape, a crack, a burn, or a damp patch that has softened the keratin. An overly damp enclosure is the commonest predisposing condition, which is one more reason for the dry surface on page 8.

SIGNS, AND WHY THIS IS NOT COSMETIC

Soft, discolored or foul-smelling patches. Pitting, flaking or lifting scutes with damp tissue underneath. A discharge or an unpleasant smell. Deep ulcerations can become life-threatening if not treated aggressively, and shell rot that reaches the bloodstream causes septicemia, which is a genuine emergency. Diagnosis involves microscopic analysis and culture to identify the organism, because a bacterial and a fungal infection need different drugs.

Treatment is cleaning and debridement, medication chosen for the organism, and correcting the husbandry that allowed it. Do not treat a suspected shell infection with a household antiseptic and hope: superficial treatment over an established infection seals it in.

Shell trauma

The commonest cause of shell trauma in a pet tortoise is **a dog in the same household**. A dog does not need to be aggressive: chewing a tortoise like a toy produces punctures and crushing injuries, and a shell is bone with a blood and nerve supply, so it hurts and it bleeds. Any damage can lead to life-threatening bone infection, and severe injuries expose internal organs.

- **Never leave a tortoise and a dog in the same space unsupervised**, including a garden, including a dog that has always ignored it.
- Other sources: a fall from a height, a heavy object, being stepped on, a lamp burn, or a heat mat under the animal.
- Any crack, puncture or bleeding is a same-day veterinary case. Keep the animal clean and dry, do not apply glue, epoxy, resin or tape, and do not submerge it.

Abscesses

Abscesses in tortoises present as **hard, tumor-like swellings** rather than the soft pockets a mammal forms, because the pus is thick and dry. The classic sites are the ear and the mouth: a large swelling on the side of the head just behind the eye is an aural abscess, and swellings in the roof of the mouth also occur. They are frequently related to vitamin A deficiency (page 22). Treatment is surgical drainage or removal, flushing with a medicated solution, culture to identify the organism, and antibiotics. They do not resolve without intervention and they do not drain on their own.

THE MONTHLY SHELL CHECK

Once a month, with the animal on your lap over a towel: look at the whole carapace under good light for soft spots, discoloration, lifting scutes and cracks; turn it over briefly and check the plastron the same way, especially where it meets the substrate; check both sides of the head behind the eyes for swelling; and press gently at the shell margins for give. It takes two minutes and it catches all three conditions on this page early, when they are all treatable.

Parasites, the fecal test & quarantine

Fairly common, mostly invisible, and the reason every new tortoise is screened.

What they carry

Roundworms and protozoa are common gastrointestinal parasites in tortoises, particularly in wild-caught animals, and in many cases a parasitic infection causes no clinical signs at all. Severe infections cause diarrhea, weight loss and failure to thrive. Oxyurid (pinworm) burdens have been found to correlate with calcium deficiency and metabolic bone disease as well as with kidney disease, mainly in juveniles, which is one more reason not to treat a parasite load as a separate issue from husbandry.

HEXAMITA PARVA, THE ONE SPECIFIC TO THIS SPECIES

Hexamita parva is a highly damaging parasite of the renal and urinary system and has been identified frequently in *Testudo horsfieldii*. Untreated it causes rapid weight loss, dehydration and terminal renal failure. Treatment is oral metronidazole, reported at 260 mg/kg, and it needs a vet: the dose is calculated on body weight and the diagnosis needs a urine or fecal examination looking for the right organism. A Russian tortoise losing weight steadily with no obvious cause deserves this specifically ruled out.

The fecal test

Every newly acquired tortoise gets a fecal test, whether or not it has symptoms. Collect a fresh dropping, keep it in a sealed bag in the fridge, and take it to the vet within 24 hours; a sample from a soak is fine. Repeat annually thereafter, and repeat immediately for any unexplained weight loss or change in droppings. Deworming is prescribed against what is actually found rather than given routinely, because blanket dosing selects for resistance and some anthelmintics are hard on a debilitated animal.

Quarantine, in full

RULE	WHY
A separate room	Not a second enclosure in the same room. Herpesvirus and mycoplasma travel on air, hands and equipment. Page 19
Months, not weeks	Three to six months is the figure most tortoise keepers use, and six is not excessive for an animal that will live forty years. Viral carriers can look well for a long time
Separate everything	Dishes, hides, thermometers, tools, substrate, and a separate pair of hands or a wash and a change of clothes between
Established animals first, the new one last	Every day, without exception
A vet visit and a fecal test during it	The point of the quarantine period is to use it. Ask specifically about screening for herpesvirus if you keep other tortoises
Weigh weekly	A new animal losing weight in quarantine is the earliest signal available. Page 15

WILD-CAUGHT ANIMALS, AND WHAT "CAPTIVE BRED" OFTEN MEANS

Large numbers of wild-caught Russian tortoises have entered the trade labeled as captive-bred or ranched, and it is likely that a substantial proportion of animals sold as captive-bred are actually wild-caught. Imported animals are frequently stressed, dehydrated, carrying injuries and carrying disease. A high parasite burden and a poor start are the norm rather than the exception. Buy from a breeder who can tell you the animal's hatch date, and expect to pay more for it.

Vitamin A deficiency, cloacoliths & beak overgrowth

Three conditions from three different corners that show up in the same tortoise.

Hypovitaminosis A

Vitamin A deficiency comes from an inappropriate diet, and the two that cause it are a diet based on iceberg lettuce and one containing meat. Its signs are easy to attribute elsewhere.

SIGNS

Lack of appetite. Lethargy. Swelling of the eyelids, often with a pus-like discharge, so that the eyes are partly or fully closed. Swelling of the ear from an aural abscess, seen as a bulge on the side of the head behind the eye. Respiratory infections, which are frequently secondary to it. Kidney failure in later stages. Treatment is veterinary vitamin A, sometimes by injection, plus a corrected diet. Do not self-dose vitamin A: it is fat-soluble and an overdose causes skin sloughing.

Prevention is the diet on pages 11 and 12: dark leafy greens, weeds and small amounts of orange-fleshed vegetables all carry the carotenoids a tortoise converts to vitamin A. A tortoise eating dandelion and plantain will not be deficient.

Cloacoliths and urate blockage

STRAINING WHILE VOCALIZING IS A TRUE EMERGENCY

A cloacolith is a blockage formed by an accumulation of urates, the crystalline portion of a reptile's urine, which harden into a stone in the cloaca. A tortoise straining and making noise while doing it is a genuine emergency: this is one of very few sounds a tortoise makes and it means obstruction. Treatment is sedation, pain relief, and breaking down and flushing the blockage with forceps. Do not attempt to remove anything yourself.

The cause is nearly always chronic dehydration, which is why the weekly soak and the always-available water dish on page 8 matter more than they look. Thick, gritty urates are the warning that comes before the blockage. Page 24.

Beak and nail overgrowth

Beak and claws grow continuously and are worn down by biting tough food and walking on abrasive ground. On soft supermarket greens and a substrate of nothing but soil, both overgrow. An overgrown beak eventually stops the animal biting properly, at which point it loses weight, and a badly overgrown one forces the jaw out of alignment permanently.

- **Prevention:** coarse fibrous weeds and hay rather than soft leaves, a permanent cuttlebone, and flat stones or paving to walk over.
- **Other causes exist**, including metabolic bone disease and a past jaw injury, so a beak that keeps regrowing badly is a symptom rather than a grooming job.
- **Trimming is a veterinary procedure**, done with a rotary tool. Do not use clippers, which crack the beak.

SHEDDING, WHICH TORTOISES DO DIFFERENTLY

A tortoise does not shed in one piece. Skin flakes from the neck and legs continuously, and scutes may shed thin translucent layers as the shell grows. Both are normal. Skin retained in tight bands around a limb means dehydration; scutes lifting to reveal soft damp tissue is page 20. Never peel a scute.

Brumation: the decision and the protocol

Programmed into the species, optional indoors, and dangerous done casually.

Russian tortoises are physiologically programmed to brumate. In their native range they are active for only about three months of the year, spending the rest underground in burrows 20 to 50 cm (8 to 20 in) deep where the soil sits at 6 to 10°C (43 to 50°F). Captive tortoises keep the drive, and many slow down, refuse food and dig in as autumn arrives whatever the enclosure is doing.

THE GENUINE CHOICE INDOOR KEEPING GIVES YOU

By maintaining full lighting, heat and feeding year-round, an indoor keeper can keep a Russian tortoise active through winter rather than brumating it. This is a real option that many species do not offer, and it is a perfectly legitimate way to keep this animal. Wild-caught adults in particular may refuse food or show real stress if prevented from brumating, so the choice is easier with a captive-bred animal, and the honest position is that both routes are defensible and that the wrong route is a half-hearted one.

Do not brumate a tortoise that is

- Underweight, or has lost weight this year
- Under about two years old, or newly acquired
- Showing any sign of respiratory infection, however mild
- Carrying an untreated parasite load
- Recovering from any illness or injury
- Not health-checked by a vet in the last few months

A pre-brumation veterinary health check is worth doing before a first brumation and after any year with a health problem in it. A tortoise that goes into brumation sick does not come out.

The protocol, if you say yes

1. **Weigh and record**, and check the shell, eyes, nose and mouth. This is the baseline every later weighing is measured against.
2. **Stop feeding, two to four weeks before** the temperature comes down, while heat and light stay on. The gut has to be completely empty: food left fermenting in a cold tortoise is a route to serious illness. Keep offering water and soaking through this period.
3. **Wind the temperature and photoperiod down gradually** over a week or two, rather than moving the animal straight into the cold.
4. **Hold it at 40 to 55°F (4 to 13°C)** for 2 to 4 months, in a ventilated box of dry substrate in a place that stays reliably in that band. A dedicated refrigerator with a thermometer and a daily door-opening for air exchange is the method most experienced keepers now use, because a garage or a shed swings with the weather.
5. **Check weekly.** Weigh the animal. A loss of more than roughly 1% of body weight a month, or any waking, urination, discharge or discomfort, ends the brumation early.
6. **Warm up gradually**, over a few days, then straight back to full basking temperature and UVB.
7. **Soak immediately on waking**, and daily for the first week. Rehydration comes before food. Expect the first meal to be refused and the first droppings to take several days.

TWO THINGS THAT KILL BRUMATING TORTOISES

Food left in the gut. The fasting period is not optional and it is not a suggestion. **Temperature that drifts.** Too warm and the animal burns reserves without being properly dormant; below freezing and it suffers permanent eye and tissue damage or dies. A brumation with no thermometer on it is a gamble.

Reading droppings, urates & hydration

A reptile passes two things, and the second one is the more useful.

A tortoise produces **feces**, the brown or dark green solid, and **urates**, a separate white or off-white portion that is the crystalline waste a reptile excretes instead of liquid urine. It also passes clear liquid urine. Each tells you something different, and the urates in particular are the best hydration gauge an owner has. The weekly soak on page 8 is when most tortoises pass both, which makes it the easiest time to look.

WHAT YOU SEE	NORMAL	WORTH ACTING ON
Feces	Formed, firm, brown to dark green, often with visible plant fiber. Every one to three days on a normal diet	Watery or loose persistently, which usually means too much fruit or watery greens, and can mean parasites or infection. Very small and hard means dehydration or too little food. Page 21
Urates	Soft, white, toothpaste-like. Passed with or shortly after feces or during a soak	Thick, pasty, gritty or chalky means dehydration. Yellow or orange suggests concentration or liver involvement. Hard lumps are the beginning of the blockage on page 22, and pink or red means blood
Urine	Clear liquid, colorless	Cloudy, dark, or with grit in it
Whole undigested food	Some visible fiber is normal	Recognizable whole pieces passing through means the gut is not working. Vet
Nothing at all	A few days between droppings is normal, longer in cool weather	More than about a week outside brumation, particularly with a reduced appetite or a bloated feel, needs a vet

GRITTY URATES ARE THE SIGN TO CATCH

Thick, sandy, chalky urates mean the animal is running chronically dry. It is the earliest and easiest warning this species gives, it precedes the urate blockage that is a true emergency, and it is fixed with the two cheapest interventions in this book: more soaking and a genuinely moist hide. Look at the soak water every week.

Hydration

A tortoise from an arid environment is very good at conserving water, which means dehydration is well advanced before it is visible. The signs are sunken eyes, loose or wrinkled skin at the limb sockets, thick urates, reduced appetite, and lethargy. The interventions are all routine rather than emergency:

- **A shallow water dish, sunk level with the substrate, changed daily.** Many tortoises drink far more than their keepers expect once the dish is easy to walk into.
- **A weekly soak** of 20 to 30 minutes in shallow warm water reaching the bottom of the shell, more often for hatchlings, newly acquired animals, and any tortoise with thick urates.
- **A moist hide**, remoistened every few days. Page 8.
- **Water-containing food.** Fresh weeds and greens carry real moisture; dry hay alone does not.

SOAKING IS ALSO A HEALTH CHECK

A tortoise in a soak is level, still, and easy to look at from every angle. Use the twenty minutes: check the eyes and nose, look at the shell top and bottom, feel the limb sockets, and inspect what ends up in the water. It is the single most efficient routine in this book.

The law & the four-inch rule

The species is legal almost everywhere. The rule everyone quotes is not what they think it is.

Of the species we have checked state by state, the Russian tortoise has the fewest restrictions in the United States. **No state bans it. One state requires a permit. Hawaii, which bans almost everything, expressly allows it.** The rule that actually shapes how these animals are bought and sold is federal, says nothing about the species, and is misquoted more often than any other rule in the pet trade.

WHAT THE FOUR-INCH RULE ACTUALLY SAYS

It is 21 CFR 1240.62, an FDA regulation from 1975 written to reduce *Salmonella* transmission to small children. Its operative sentence is that viable turtle eggs and live turtles with a carapace length of less than 4 inches **shall not be sold, held for sale, or offered for any other type of commercial or public distribution.** Sold, held for sale, offered for distribution. **Nothing about owning one,** and nothing about the species: it catches a hatchling Russian tortoise for the same reason it catches a hatchling red-eared slider.

Then comes the part almost nobody quotes. Subsection (c) lists exceptions, and the second covers the sale, holding for sale and distribution of live turtles and viable turtle eggs **not in connection with a business.** A private keeper rehoming a hatchling is not a business; a breeder selling them is. That clause is why hatchlings move hand to hand through private keepers while shops stock only animals past the four-inch line, and it is the real reason a Russian tortoise you buy is usually already a few years old.

WHERE	THE RULE	IN PRACTICE
New Jersey	Not on the exempt species list, so the catch-all permit provision applies	A personal possession permit from Fish and Wildlife. The only one in the country
Hawaii	The genus <i>Testudo</i> is on the List of Conditionally Approved Animals	Allowed, subject to Department of Agriculture conditions. Rare for Hawaii, which works from a closed list where not being named is enough to be banned
Montana	Cleared by family, with the federal size floor written into the statute	Legal above four inches of shell
Rhode Island	All exotic turtles allowed without a permit except named exceptions, which do not include this species	Legal, while the state's own native turtles cannot be kept at all
Everywhere else	No state names the species	Legal. Your city ordinance is the only thing left to check

CONSERVATION STATUS, AND WHY IT SHOULD MATTER TO A BUYER

Testudo horsfieldii is Vulnerable on the IUCN Red List and CITES-listed, and large numbers of wild-caught animals have entered the trade labeled captive-bred or ranched. A cheap adult at an expo is very often a wild-caught import, with the parasite load and disease risk that carries (page 21). Captive-bred costs more and does not draw on a wild population.

Rules change, and this page is a summary rather than legal advice. Confirm your own state and city before buying, and check import rules before crossing a border with a CITES-listed animal.

Setup checklist & targets

Print this one. Audit an existing setup against it, or buy against it.

Enclosure

- ☐ Open-topped tortoise table, not a glass tank
- ☐ Floor at least 4×2 ft (120×60 cm); 8×4 ft (240×120 cm) is the target
- ☐ Walls 12 in (30 cm) or more, nothing climbable against them
- ☐ Substrate 4 to 6 in (10 to 15 cm), topsoil and coir mix, damp at depth
- ☐ Two hides at least: one warm and dry, one cool and moist
- ☐ Shallow water dish sunk level, changed daily
- ☐ Flat stone or tile as a feeding surface
- ☐ Cuttlebone left in permanently

Heat and light

- ☐ Halogen basking lamp, surface reading 95 to 100°F (35 to 38°C)
- ☐ Cool end genuinely in the 70s°F (21 to 26°C)
- ☐ No heat mat anywhere under the animal
- ☐ Dimming thermostat or lamp dimmer on the basking bulb
- ☐ Linear T5 HO UVB tube, a third to half the enclosure length
- ☐ UVI 3.0 to 4.0 at the basking site; mounting distance checked
- ☐ Bulb replacement date written on the bulb and on page 35
- ☐ Timer set for 12 hours on, 12 off

Instruments, food and care

- ☐ Infrared temperature gun for the basking surface
- ☐ Digital probe thermometer at the cool end, plus a hygrometer
- ☐ Digital kitchen scale reading in grams
- ☐ Calcium without D3, calcium with D3, herbivore multivitamin
- ☐ Weeds or a forage seed mix growing, plus grass hay always available
- ☐ Weekly soak established, 20 to 30 minutes
- ☐ Reptile vet with tortoise experience, number on page 27
- ☐ Fecal test booked for the first month

Targets at a glance

Basking surface	95 to 100°F (35 to 38°C)
Cool end and night	70s°F (21 to 26°C); night to 60°F (16°C) is fine
UVB	UVI 3.0 to 4.0 at the basking site, bulb replaced every 12 months
Humidity	Dry ventilated surface with a moist hide. Page 8
Diet	Weeds and hay, low protein, no fruit; adult portion the size of the shell
Vet	Fecal test on arrival and annually, plus a yearly wellness exam

Emergency & quick targets card

Fill this in today. Cut it out or photograph it and keep it on your phone.

Reptile vet _____

Phone _____ Hours _____

After-hours / exotics emergency clinic _____

Phone _____ Drive time _____

Backup carer _____ Phone _____

Long-term carer, if I cannot keep it _____

Tortoise name _____ Hatch or purchase date _____

Sex _____ Baseline weight (g) _____

Go now, do not wait

- Open-mouth breathing, gasping, neck extended
- Bubbles or mucus at nose or mouth
- Straining while making noise. Page 22
- A female straining or unable to lay. Page 16
- Any shell crack, puncture or bleeding, especially after a dog
- Prolapse from the vent
- Collapse or unresponsiveness after a hot day
- Cannot right itself, or cannot lift its body to walk

WHILE YOU TRAVEL

A ventilated box lined with damp paper towel, no substrate to shift around, and no water dish to spill. Keep it warm, around 75 to 80°F (24 to 27°C), and dark and quiet. In cold weather use a wrapped hot water bottle beside the box, never touching the animal. Take a photograph of the enclosure and the last droppings, and bring the UVB bulb's age, the basking temperature reading, and the current weight.

Targets, restated

Enclosure floor	4×2 ft (120×60 cm) minimum, 8×4 ft (240×120 cm) target, open top
Basking surface	95 to 100°F (35 to 38°C), measured on the surface
Cool end / night	70s°F (21 to 26°C); night to 60°F (16°C). Act if sustained below 60°F
UVB	UVI 3.0 to 4.0; T5 HO tube replaced every 12 months
Substrate	4 to 6 in (10 to 15 cm), damp at depth, dry at the surface
Soak	20 to 30 minutes weekly, shallow warm water
Brumation	40 to 55°F (4 to 13°C), 2 to 4 months, only if healthy. Page 23
UVB bulb fitted _____	Replace by _____
Last fecal test _____	Last wellness exam _____

Budget & shopping list

Itemized, and the lines add up. US retail, September 2026.

One-time setup

ITEM	LOW	CONSIDERED
Tortoise table, bought or built, 4×2 ft or larger	\$125	\$250
T5 HO UVB fixture and bulb	\$60	\$130
Halogen basking lamp, fixture and a dimming thermostat	\$50	\$105
Timer	\$10	\$20
Infrared temperature gun	\$18	\$30
Digital probe thermometer, hygrometer and substrate fill	\$40	\$90
Hides, cork, flat stones and a dig box	\$30	\$70
Water dish, feeding tile and a digital gram scale	\$24	\$50
Calcium, calcium with D3, multivitamin, cuttlebone	\$15	\$30
Equipment subtotal	\$372	\$775
The tortoise, hatchling captive-bred to adult	\$50	\$400
First vet exam plus a fecal test	\$75	\$200
Total to get started	\$497	\$1,375

Ongoing, per year

ITEM	LOW	CONSIDERED
Greens, weeds and grass hay	\$140	\$280
Supplements	\$20	\$35
Substrate replacement	\$40	\$90
UVB bulb, replaced annually	\$45	\$90
Electricity for heat and light	\$60	\$120
Annual vet wellness check with a fecal test	\$60	\$100
Annual total	\$365	\$715
Which is, per month	\$30	\$60

THE FORTY-YEAR NUMBER

At \$365 to \$715 a year, a Russian tortoise costs roughly **\$15,000 to \$30,000 over a forty-year life**, before a single illness. Sick or emergency care with tests and medication commonly runs \$150 to \$300 or more per episode, and this animal will have episodes. Nobody budgets a tortoise this way and everybody should: the purchase price is the least significant number on this page.

Electricity moves most: a cold house costs meaningfully more to keep a basking spot in, and a tortoise outdoors for half the year costs almost nothing to heat. Save on the table, which is a plywood box anyone can build, on substrate bought as topsoil rather than as a reptile product, and on food, which grows free in a lawn (page 12). Do not save on the UVB fixture, the thermostat, or the vet: a coil bulb costs a fraction of a T5 kit and produces the deformity on page 18.

First 30 days checklist

In order. Most of week one is letting it settle and watching what comes out.

Before it arrives

- ☐ Enclosure built, filled, lit and running for several days
- ☐ Basking surface measured with the infrared gun and adjusted to 95 to 100°F
- ☐ Cool end confirmed genuinely cool with a probe thermometer
- ☐ UVB mounted at the right distance, date written on the bulb
- ☐ Reptile vet found and the number written down
- ☐ Quarantine room ready if you already keep reptiles. Page 21

Days 1 to 7

- ☐ Weigh on arrival and write the number down. This is the baseline for the next forty years
- ☐ Handle only to move it in and to soak it. Otherwise leave it alone
- ☐ Soak daily this week, 20 minutes, shallow warm water. Newly moved tortoises are usually dehydrated
- ☐ Offer food daily and do not worry if it is refused for the first few days
- ☐ Watch for the first droppings and urates, and keep a fresh sample for the vet
- ☐ Check the basking temperature again with the animal actually in the enclosure

Days 8 to 14

- ☐ Veterinary health check plus a fecal test, symptoms or not. Page 21
- ☐ Ask the vet to confirm sex if the animal is large enough, and to review the supplement schedule
- ☐ Weigh again. A newly moved tortoise often dips slightly, then recovers
- ☐ Switch the diet gradually toward weeds and hay if it arrived on something else. Page 11

Days 15 to 30

- ☐ Weekly soak established, and the weekly weigh-in logged. Page 34
- ☐ Scatter feeding started, in different places each day. Page 14
- ☐ Weeds identified in your own area, or a forage tray sown. Page 12
- ☐ First monthly shell check done. Page 20
- ☐ First supervised outdoor session if the weather allows. Page 9
- ☐ Long-term carer named and written on page 27

WHAT NORMAL LOOKS LIKE BY DAY 30

Out and walking the enclosure in the morning, basking for a stretch, eating within a few hours of food appearing, digging into the substrate or a hide by late afternoon, and producing formed droppings with soft white urates every couple of days. A tortoise that hides constantly, refuses food for weeks, or has not passed anything is not settling in, it is unwell. Page 17.

Symptom quick reference

What you see, what it might be, and how fast to move.

WHAT YOU SEE	MOST LIKELY CAUSES	ACT
Open-mouth breathing, gasping, neck extended	Respiratory infection. Page 19	Now
Bubbles or mucus at nose or mouth	Respiratory infection, mycoplasma, vitamin A deficiency	Now
Straining and making noise	Cloacolith or urate blockage. Page 22	Now
Female straining, digging then stopping, off food	Egg binding. Page 16	Now
Shell cracked, punctured or bleeding	Dog, fall, crush. Page 20	Now
Prolapse from the vent	Straining, blockage, egg binding, parasites	Now
Collapsed or unresponsive after a hot day	Overheating. Move to shade, cool gradually, then vet	Now
Cannot lift its body to walk, or wobbles	Metabolic bone disease. Page 18	Same day
Shell soft or flexible past hatchling size	Metabolic bone disease	Same day
Swollen eyelids, eyes closed, pus-like discharge	Vitamin A deficiency. Page 22	Within days
Hard swelling behind the eye	Aural abscess. Page 20	Within days
Soft, discolored or smelly patches on the shell	Shell rot. Page 20	Within days
Persistent runny nose	Never just a cold. Infection, mycoplasma, or vitamin A	Within days
Steady weight loss	Parasites, including <i>Hexamita</i> , dental or beak problems, kidney disease. Page 21	Within days
Thick, gritty, chalky urates	Dehydration heading toward a blockage. Page 24	Soak now, vet if it persists
Watery droppings	Too much fruit or watery greens, parasites, infection	Correct diet, then vet
Nothing passed for over a week outside brumation	Impaction, blockage, dehydration, too cold	Within days
Raised, cone-shaped scutes appearing	Pyramiding. Reduce food, add water and exercise. Page 18	Change husbandry now
Overgrown beak, difficulty biting	Soft diet, MBD, past injury. Page 22	Within weeks
Skin retained in tight bands on a limb	Dehydration. Soak daily and check the moist hide	Within days
Refusing food in autumn, digging in, slowing down	Brumation instinct. Page 23	Decide, do not drift
Pacing a wall for hours	Glass sides, or an enclosure too small. Pages 5 and 14	Change the enclosure
Ramming or biting another tortoise	Male territoriality. Separate. Page 10	Now

Daily, weekly & seasonal routine

Ten minutes a day, half an hour a week, and one decision each autumn.

Every day

- ☐ Fresh food, scattered in new places, on a tile or stone
- ☐ Water dish emptied, rinsed and refilled
- ☐ Remove yesterday's uneaten greens and any droppings
- ☐ Look at the tortoise properly: gait, eyes, nose, and whether it basked
- ☐ Check the basking surface reading and the cool end
- ☐ Damp the moist hide if it has dried out

Every week

- ☐ Soak, 20 to 30 minutes, and use it as a health check. Page 24
- ☐ Weigh and log the number. Page 34
- ☐ Top up hay and refresh the forage trays
- ☐ Turn and refresh the substrate surface; replace any soiled patch
- ☐ Rearrange non-critical decor, leaving the hides where they are
- ☐ Walk the perimeter of any outdoor pen looking for digging

Every month

- ☐ Shell check, top and bottom, under good light. Page 20
- ☐ Check the shell margin for the ridging that starts pyramiding. Page 15
- ☐ Full substrate turn or partial replacement, and a wash of dishes and hides
- ☐ Review the weight log as a trend rather than as this month's number
- ☐ Check the UVB bulb's replacement date. Page 35

Seasonal

WHEN	WHAT CHANGES
Spring	Weeds return. Start picking, sow forage trays, and begin outdoor sessions once daytime temperatures allow. A tortoise coming out of brumation is soaked daily for the first week
Summer	The dangerous season. Check the enclosure and the pen at the hottest part of the afternoon, confirm permanent shade, and never leave a tortoise in a conservatory, greenhouse, car or unshaded pen. Page 6
Autumn	The brumation decision, made deliberately rather than drifted into. If yes, the fasting period starts now. If no, keep heat, light and feeding fully on and expect the tortoise to slow down anyway. Page 23
Winter	Either the animal is in brumation and checked weekly, or it is fully active and the enclosure is fighting a cold house. Watch for drafts, and expect the electricity line on page 28 to rise
Yearly	UVB bulb replaced on the date, not on failure. Vet wellness exam with a fecal test. Substrate fully replaced. Enclosure and pen checked for digging damage and rot

Power outages, travel & transport

A reptile depends on an outlet for the one thing it cannot make: body heat.

A healthy adult Russian tortoise tolerates one ordinary cool night without harm. The risk is duration, not the single low reading, and this species has more margin than almost any other reptile kept: it **tolerates 60°F (16°C) comfortably**, and the point to act is a sustained period below 60°F outside an intentional brumation. Pushed meaningfully below its preferred range for more than a day or two, a reptile becomes more susceptible to bacterial infection and stops eating, which is the textbook setup for the respiratory infection on page 19.

During an outage

- **Move the enclosure to the warmest interior room**, away from windows and exterior doors.
- **Drape thick towels or a blanket over the enclosure, leaving a vent gap**, to hold in the heat that is left.
- **A wrapped hot water bottle or a commercial hand warmer beside the animal**, never touching it directly and never underneath it.
- **Stop feeding** if the enclosure is genuinely cold. A reptile cannot digest below its working temperature, and food sitting undigested in a cold gut is a real problem. This is the opposite of the rule for a mammal or a bird.
- **Keep water available**, and keep soaking on the normal schedule with warm water heated another way.
- **No candles, gas stoves or open flames near the enclosure**. The fire and carbon monoxide risk outweighs the cold, which this species handles well.

Outage kit, packed before you need it: two or three hand warmers or a hot water bottle, two thick towels, a battery or wind-up thermometer, a ventilated tub sized for the animal, and your reptile vet and 24-hour exotics numbers written on paper rather than saved on a phone that may die.

Travel and being away

TRIP LENGTH	WHAT A HEALTHY ADULT NEEDS
1 to 2 nights	Often fine alone. Lights on timers, thermostat running, water dish full, and a generous scatter of greens and hay before you leave
3 to 7 nights	A sitter every 1 to 2 days for fresh food and water, a look at the droppings, and a temperature check
Over a week	A daily or every-other-day sitter who has done a practice visit, or boarding with a reptile-experienced vet or shop. Never a general boarding kennel
A hatchling or juvenile	Shift every row one tier stricter. Young tortoises are soaked more often and eat daily

TRANSPORT

A ventilated box lined with damp paper towel, no loose substrate, no water dish. Warm, around 75 to 80°F (24 to 27°C), dark and quiet, and belted into the car rather than loose on a seat. Never in the trunk, never in a parked car in any weather, and never for longer than necessary. For a long move, the tortoise travels with you, and its enclosure is set up and running at the far end before it comes out of the box.

THE FORTY-YEAR VERSION OF THIS PAGE

A tortoise outlives jobs, houses and relationships. The plan that actually matters is not the weekend one, it is who takes this animal in thirty years. Name someone, tell them, and write it on page 27. Tortoise rescues exist because this conversation usually happens too late.

Pet-sitter sheet

Fill it in and leave it by the enclosure. Assume the sitter has never kept a reptile.

Tortoise name _____ Sex _____

Owner _____ Phone _____

Reptile vet _____ Phone _____

Emergency clinic _____ Phone _____

Backup contact _____ Phone _____

Normal weight (g) _____ Visit every _____

Each visit, please

- | | |
|---|--|
| ☐ Empty, rinse and refill the water dish | ☐ Check the two thermometers and write down what they say |
| ☐ Scatter fresh greens on the stone, in a couple of places | ☐ Damp the moss in the humid hide if it feels dry |
| ☐ Remove yesterday's uneaten food and any droppings | ☐ Look at the tortoise walking. Text us a photo |

PLEASE NEVER

Turn the lamps off, or unplug anything. The lights are on timers and the tortoise needs them.
 Move the tortoise outside, or take it out of the enclosure except as we have shown you.
 Let a dog or cat near it, even in the same room.
 Feed it fruit, meat, bread, dog or cat food, or anything from the never list below.
 Put it on its back, or lift it by a leg.

Never feed

Fruit, any meat or animal food, dog or cat food, bread, pasta, cereal, iceberg lettuce, avocado, rhubarb, or any plant from a garden. Only what is in the box in the fridge that we have labeled.

Call us if you see

- | | |
|---|---|
| • Breathing with its mouth open, or bubbles at the nose | • Food untouched across two visits |
| • Not moving, or not coming out to bask at all | • A lamp that has stopped working |
| • A crack or bleeding anywhere on the shell | • Anything that just seems different from last time |

Where the food is _____

Which lamps are on timers, and when _____

Anything else _____

Owner log

Weigh weekly at the soak, or monthly once the animal is a settled adult, and always at the same point in the routine. Photocopy the page before you start; forty years is a lot of rows. Sudden weight loss in a tortoise is always significant, because these animals lose weight slowly by nature, and a chart is what turns "it looks a bit smaller" into a number a vet can use.

Equipment, supplement & vet log

The bulb date is the most important line on this page.

Equipment

ITEM	FITTED	REPLACE BY	NOTES
UVB tube (12 months)			
Basking bulb			
Substrate fully replaced			
Thermostat / timer checked			

Supplements & soaks

WEEK	CALCIUM (NO D3)	CALCIUM + D3	MULTIVITAMIN	SOAK
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐
	☐	☐	☐	☐

Veterinary

DATE	REASON	WEIGHT (G)	FINDINGS, TREATMENT, FOLLOW-UP

Enrichment, ticked monthly

- ☐ Scatter feeding in new places
- ☐ Living forage to crop
- ☐ Decor rearranged, hides left alone
- ☐ Supervised outdoor time
- ☐ Dig area refreshed and dampened
- ☐ New terrain: stone, slope or log

REFERENCE

Glossary

Terms used earlier in this book, in the sense they were used.

Brumation	The reptile equivalent of hibernation. A period of cold dormancy, 2 to 4 months in this species, and optional indoors. Page 23
Carapace	The upper shell
Cloaca	The single opening for waste and reproduction. Cloacolith: a hardened urate blockage in it
Ferguson Zone	A classification of how much UV a species seeks out in the wild, used to match a bulb to an animal rather than buying the strongest one
Gravid	Carrying eggs. Fertile or not
Herpesvirus	A viral disease of tortoises with no cure, high contagion, and near-total mortality in unquarantined groups. Page 19
Hexamita parva	A parasite of the kidney and urinary tract found frequently in this species. Page 21
MBD	Metabolic bone disease, from a calcium and phosphorus imbalance, inadequate UVB, or both. Page 18
Mycoplasma	The bacterium behind chronic upper respiratory tract disease in tortoises: the runny nose that never fully clears
Plastron	The lower shell
Pyramiding	Raised, cone-shaped scutes from overfeeding, excess protein or chronic dehydration. Permanent. Page 18
SCL	Straight carapace length. The standard measurement, taken in a straight line rather than over the curve
Scute	One of the keratin plates covering the shell. They grow at their margins, which is why the shell is a record
SCUD	Septicemic cutaneous ulcerative disease, an older term for severe shell infection. Page 20
T5 HO	A high-output linear fluorescent tube, the correct format for reptile UVB. Page 7
Urates	The white, semi-solid crystalline waste a reptile passes instead of liquid urine. The best hydration gauge an owner has. Page 24
UVI	UV index, the measured strength of UV reaching a point. Target 3.0 to 4.0 at this species' basking site
21 CFR 1240.62	The FDA four-inch rule. It restricts commercial sale, not ownership. Page 25

REFERENCE

Sources, version history & about

Where the numbers in this book came from.

Veterinary and specialist sources

SOURCE	WHAT IT SUPPLIED
VCA Animal Hospitals, <i>Common Diseases of Tortoises</i>	Metabolic bone disease causes and signs, pyramiding, hypovitaminosis A, respiratory tract infection signs and treatment, abscesses, shell infections and SCUD, shell trauma, cloacoliths, and parasites
Tortoise Trust, <i>Captive Care of the Russian Tortoise</i>	Maximum verified size, sexual dimorphism, the burrow temperature figures, the substrate depth for a pallet, the buried-barrier depth, the 80-plant wild diet and the ninety-days-a-year feeding pattern, the low protein and no fruit rule, <i>Hexamita parva</i> with its metronidazole dose, herpesvirus risk, and the wild-caught trade
ReptiFiles, <i>Russian Tortoise Care Sheet</i>	The 8×4 ft enclosure figure and per-additional-animal space, the UVB bulb, mounting distances and the UVI 3.0 to 4.0 target, the 40 to 75% humidity position, and the substrate depth
Merck / MSD Veterinary Manual	The preferred optimal temperature zone framing behind the cold-tolerance figures on pages 6 and 32
FDA, 21 CFR 1240.62, and state statutes and regulations	The four-inch rule and its business exception, New Jersey's permit provision, Hawaii's List of Conditionally Approved Animals, Montana's statute and Rhode Island's exceptions. Page 25
IUCN Red List and CITES	The Vulnerable listing and the trade status

Beastly Facts guides drawn on: Russian tortoise enclosure setup, health issues, handling, enrichment, cost and legal, plus the complete UVB lighting guide, the T5 versus compact comparison, and the reptile emergency plan. Where the site and a specialist source disagreed, both figures are printed and the disagreement is named.

Version history

VERSION	DATE	TEMPLATE	WHAT CHANGED
1.0	Aug 2026	pre-t3	First edition, 21 pages on the earlier layout
2.0	Sep 2026	t3	Full rebuild on the current template. New pages for outdoor housing, weeds and grazing, sexing and body condition, eggs and egg binding, brumation, droppings and hydration, the legal position, and six further health pages. Enclosure size, humidity and diet all rewritten around specialist sources, with disagreements printed rather than averaged. Budget itemized so the lines sum

Beastly Facts publishes fact-checked care references for keepers who would rather read one accurate thing than ten contradictory ones. This is a husbandry reference, not veterinary advice, and the legal page is a summary rather than legal advice.

Russian Tortoise Care Package · Version 2.0 · Template t3 · September 2026 · 37 pages · For personal use by the purchaser. Please don't redistribute.